# Splunk SIEM Home Lab

## Complete Setup Documentation & Portfolio Reference

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## 1. Lab Architecture

| Component | Details |
|---|---|
| Host Machine | Windows 11 (your personal computer) |
| Virtualization | Oracle VirtualBox 7.2.10 |
| VM OS | Ubuntu Server 22.04 LTS (Jammy Jellyfish) |
| SIEM Platform | Splunk Enterprise 10.4.0 |
| Log Agent | Splunk Universal Forwarder 10.4.0 |
| VM RAM | 4096 MB (4 GB) |
| VM CPU | 2 cores |
| VM Disk | 50 GB (dynamically allocated) |
| Network Mode | Bridged Adapter (Realtek Wi-Fi 6 NIC) |
| | |

## 2. Network Configuration

| Setting | Value |
|---|---|
| VM IP Address | 192.168.1.94 |
| Splunk Web UI | http://192.168.1.94:8000 |
| Forwarder Port | 9997 (TCP) |
| SSH Access | ssh crystaleemtz@192.168.1.94 |
| Splunk Mgmt Port | 8089 |

## 3. Credentials & Accounts

Three separate accounts exist by design — each belongs to a different system:

| Account | Purpose |
|---|---|
| Ubuntu SSH (crystaleemtz) | Log into the Linux VM via PowerShell SSH |
| Splunk Enterprise (admin) | Log into Splunk web dashboard at :8000 |
| Universal Forwarder (admin) | Manage the Windows log forwarding agent |

Note: This separation is intentional security design (least privilege). Each component has its own credentials so a compromise of one doesn't expose the others.

## 4. Daily Startup & Shutdown Procedure

### Starting the Lab

- Start the VM in VirtualBox (click Start)
- Wait for Ubuntu to fully boot (~60 seconds)
- Open PowerShell on Windows and SSH in:
  `ssh crystaleemtz@192.168.1.94`
- Start Splunk inside the VM:
  `sudo /opt/splunk/bin/splunk start --run-as-root`
- Open browser and go to:
  `http://192.168.1.94:8000`
- Log in with Splunk admin credentials

### Shutting Down the Lab

- Always stop Splunk cleanly before shutdown (prevents index corruption):
  `sudo /opt/splunk/bin/splunk stop`
- Then shut down the VM:
  `sudo shutdown now`
- Do NOT close VirtualBox window directly while VM is running

## 5. Data Sources Configured

| Log Source | What It Captures |
|---|---|
| Windows Security Log | Login attempts, authentication events, account changes, audit events |
| Windows System Log | OS-level events, service failures, driver issues, system errors |
| Windows Application Log | Application-level errors, warnings, and informational events |

## 6. Key SPL Searches

SPL (Search Processing Language) is Splunk's query language. All searches are run in the Search & Reporting app.

### Basic Searches

- **All events:** index=*
- **All events with host filter:** index=* host=*

### Security-Focused Searches

- **Successful logins (Event 4624):** index=* EventCode=4624
- **Failed logins (Event 4625):** index=* EventCode=4625
- **Failed logins by username:** index=* EventCode=4625 | stats count by Account_Name
- **Top 10 most frequent events:** index=* | top limit=10 EventCode
- **Events by source type:** index=* | stats count by sourcetype

### Key Windows Event Codes Reference

| Event Code | Description |
|---|---|
| 4624 | Successful account login |
| 4625 | Failed account login (wrong password, locked account) |
| 4648 | Login attempt using explicit credentials |
| 4672 | Special privileges assigned to new logon |
| 4688 | New process created (useful for malware detection) |
| 4698 | Scheduled task created |
| 4720 | User account created |
| 4726 | User account deleted |
| 7045 | New service installed on system |

# 7. Troubleshooting Log

Issues encountered and resolved during setup:

| Issue | Resolution |
|---|---|
| Clipboard paste not working in VM console | Ubuntu Server has no GUI — switched to SSH from PowerShell instead |
| SSH connection timeout (10.0.2.15) | VM was on NAT mode. Changed VirtualBox network to Bridged Adapter, got new IP 192.168.1.94 |
| Guest Additions: no medium found on /dev/cdrom | Mounted using /dev/sr0 instead (Linux device name for virtual optical drive) |
| Splunk: Running as root deprecated warning | Added --run-as-root flag to start command |
| Universal Forwarder login failed (random password) | Reset via user-seed.conf file, re-seeded admin credentials |
| Forward server showing inactive | Configured receiving port 9997 in Splunk web UI under Settings > Forwarding and receiving |

# 8. Portfolio Project Writeup

Use this text for your portfolio website and LinkedIn:

> ### Splunk SIEM Home Lab
>
> Status: In Progress | Tools: Splunk Enterprise, Ubuntu Server 22.04, VirtualBox, PowerShell, SPL
>
> Built a full SIEM pipeline from scratch on a Windows host: provisioned an Ubuntu Server 22.04 VM in VirtualBox, installed Splunk Enterprise 10.4.0, and configured the Splunk Universal Forwarder to ship Windows Security, System, and Application Event Logs into the SIEM in real time. Resolved multiple real-world troubleshooting challenges including VM network mode configuration (NAT to Bridged), SSH-based remote management of a headless Linux server, optical drive device naming differences between VirtualBox and Linux (/dev/sr0 vs /dev/cdrom), and Splunk credential management via user-seed.conf. Currently collecting and searching 7,000+ real Windows events using SPL queries focused on authentication monitoring (Event Codes 4624/4625) and anomaly detection. Next phase: custom dashboards, alert rules, and log normalization demonstrations tied to CompTIA Security+ SIEM concepts

## 9. Interview Narrative

Use this when asked about projects in interviews:

"I built a home SIEM lab to get hands-on practice correlating and analyzing security logs, since that's core to SOC analyst work and I wanted experience beyond just studying for Security+.

I started by setting up VirtualBox on my Windows machine and provisioning an Ubuntu Server 22.04 LTS VM — I gave it 4 GB of RAM and 2 CPU cores, leaving enough headroom on my host machine, and a 50 GB dynamically-allocated disk so I wasn't over-committing storage upfront.

During setup I ran into a few real troubleshooting moments — for example, VirtualBox's Guest Additions weren't installed by default, which broke clipboard sharing and drag-and-drop between my host and the VM. I worked through mounting the Guest Additions ISO manually, figuring out that Linux was referencing the optical drive as /dev/sr0 rather than the expected /dev/cdrom symlink, and installing it via the kernel module build process. That's actually a good parallel to real environments — VM tooling and OS expectations don't always align cleanly, and you have to read system output carefully to diagnose where the breakdown is.

From there, I installed Splunk Enterprise on the Ubuntu VM, configured a receiving port on port 9997, and set up the Universal Forwarder on my Windows machine to forward Security, System, and Application Event Logs in real time. I now have over 7,000 real Windows events indexed and searchable. I've been writing SPL queries to track failed login attempts, identify authentication patterns, and build searches for common SOC use cases like Event Code 4625 analysis. My next phase is building custom dashboards and alert rules."

## 10. Next Steps for the Lab

- Build a failed login dashboard (EventCode 4625 over time)
- Create an alert rule that triggers when failed logins exceed a threshold
- Add log normalization field extractions as a demo
- Install Splunk TA for Windows (adds richer field parsing)
- Document SPL queries learned and save them as named searches
- Integrate with CIM Carriers in Motion app logs once backend is built
- Screenshot all dashboards for portfolio
- Record a short walkthrough video for portfolio/interviews